Convex Optimization – Project 3

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a)
$$\min \sum_{t=1}^{T-3} (\dot{j}_t - \dot{j}_t^{ref})^2$$

s.t. $\|p\|_\infty \leq p^{max}$

$$v_t = \frac{p_{t+1} - p_t}{h} \qquad t = 1, \dots, T-1$$

$$a_t = \frac{v_{t+1} - v_t}{h} \qquad t = 1, \dots, T-2$$

$$\dot{j}_t = \frac{a_{t+1} - a_t}{h} \qquad t = 1, \dots, T-3$$

✓ $p \in \mathbb{R}^T$

✓ we know p_t^{ref} so we can calculate v_t^{ref} then a_t^{ref} then $\dot{j}_t^{ref}$ (by the formulas the problem gave us)

b) The code is attached.